Multi-Gaussian Prototype Metric With Dual Optimization for Cross-Domain Few-Shot Hyperspectral Image Classification

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***Abstract*—Metric-based few-shot learning (FSL) methods have shown great potential in cross-domain hyperspectral image (HSI) classification. However, existing approaches face three key challenges: First, HSIs exhibit complex spatial–spectral structures, and cross-domain distribution shifts further exacerbate feature discrepancies among same-class samples, increasing the difficulty of extracting class-consistent features. Second, in cross-domain few-shot scenarios, although some methods incorporate pseudo-label estimation to achieve class-level alignment, their effectiveness is limited by the scarcity of labeled samples in the target domain and the uncertainty of pseudo-labels, and existing domain-adversarial strategies still primarily focus on global distribution alignment, making it difficult to effectively model class-level shared features across domains. Third, intraclass feature modeling typically relies on a single class prototype, which fails to capture the complex and long-tailed distribution of intraclass features in few-shot learning scenarios. To address these challenges, a multi-Gaussian prototype metric with dual optimization (MGPDO) is proposed for cross-domain few-shot HSI classification. Specifically, for consistent feature extraction from samples with complex structures, a multiscale feature extractor is designed to obtain features at multiple scales via parallel convolutions and pyramid pooling structures. For class-level alignment, the prototype-based cyclic domain triplet loss is proposed, integrating domain-adversarial learning with cross-domain prototype contrast. The module iteratively optimizes prototype pairs to increase similarity between positive pairs and reduce correlation between negative pairs, promoting class-level alignment and shared features extraction across domains. Furthermore, for intraclass feature modeling, a multi-Gaussian prototype metric is introduced, which assigns multiple Gaussian prototypes to each class to construct a one-to-many mapping, effectively modeling complex long-tailed feature distribution and improving classification performance. Experimental results on three standard hyperspectral datasets validate the effectiveness of the proposed method, offering a novel perspective on deep learning-based cross-domain few-shot HSI classification.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are widely applied in many fields such as geology [1], [2], agriculture [3], and biology [4] due to their rich spectral information. HSI classification, which aims to assign spectral pixels in images, has attracted increasing attention over the past few decades [5], [6], [7].

In recent years, deep learning-based HSI classification has gained increasing attention from the research community [8], [9], [10], [11], [12]. Early studies mainly focused on single-domain supervised learning frameworks, where convolutional neural networks were employed to model spectral and spatial information jointly [13], [14], [15], [16]. Li et al. [17] proposed a 3-D CNN that simultaneously captured spectral–spatial features, significantly improving classification accuracy. Zhao et al. [19] proposed a residual channel attention-based few-shot learning (FSL) method that enhances cross-domain feature representation via residual concatenation and random recalibration. Liu et al. [20] designed a central attention network tailored for pixel-level HSI classification, while the semantic segmentation of hyperspectral images (SegHSIs) [21] and multiarea target attention (MATA) [22] further improved classification accuracy and multiscale spatial encoding through attention-guided modules. Although deep learning has achieved great success in HSI classification under ideal conditions, such as sufficient labeled data and the training and testing data followed the same distribution, existing methods face three major challenges in constrained practical applications. First, hyperspectral data contain complex spatial–spectral structures [23], [24], which makes it difficult to extract consistent features effectively for the same-class samples with diverse structures. Second, the acquisition of HSI data is affected by various factors such as sensor nonlinearity, seasonal variation, and weather conditions [25], [26], leading to discrepancies in the spectral reflectance of the same class across different domains, a phenomenon known as domain shift [27]. Third, the annotation process for HSIs is time-consuming and labor-intensive, and the uncertainty of the target domain makes it difficult to obtain sufficient labeled

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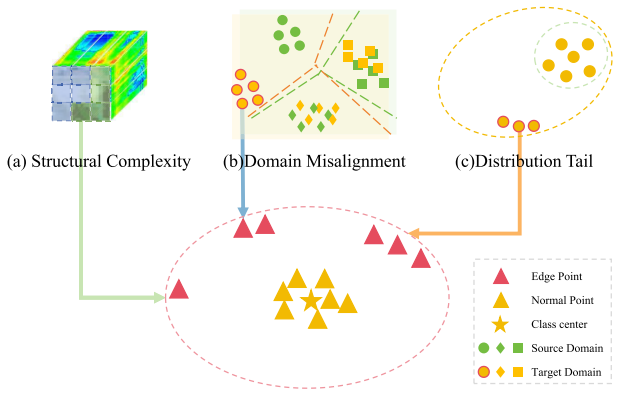


Fig. 1. Causes of edge points in cross-domain few-shot HSI classification. (a) Features of complex-structured samples tend to deviate from the class center of normal points and form edge points. (b) Class-level misalignment results in inaccurate mapping of target samples, which increases edge points. (c) In few-shot learning tasks, the limited labeled samples fail to fully cover the intraclass diversity, leading to same-class samples with low similarity residing in the tail regions and becoming edge points.

samples in the target domain for most cross-domain classification tasks, while labeled samples in the source domain are abundant [28]. Therefore, how to achieve robust feature extraction and accurate category metric under cross-domain scenarios with limited labeled samples has become a key challenge in current HSI classification.

To extract consistent features from hyperspectral data, most existing methods adopt spectral–spatial convolutional architectures [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. However, due to their shallow architectures and fixed-scale operations, these methods typically struggle to model the complex spatial dependencies in hyperspectral data. This challenge is particularly pronounced in samples containing heterogeneous pixels, especially those with complex spatial structures or located near class boundaries, where interference from heterogeneous pixels hinders the extraction of consistent features for the same-class samples. This causes the features of complexly structured samples to deviate from the class center and form edge points, as illustrated in Fig. 1(a). These samples are easily confused with neighboring classes and are prone to misclassification.

In addition to feature extraction challenges, domain shift in cross-domain tasks amplifies the difference between source and target feature distributions, reducing the transferability of discriminative features and thus limiting the model's generalization ability on the target domain [31]. To mitigate the impact of domain shift, researchers have proposed domain adaptation (DA) mechanisms [32], [33]. These methods aim to align the feature spaces of the source and target domains to mitigate domain shift, enabling the model to learn domain-invariant features and improve its generalization ability on the target domain. Some DA methods are developed based on statistical techniques. Maximum mean discrepancy (MMD) can be used to measure and minimize the distribution discrepancy between the source and target domains [34], [35]. Other methods employ an adversarial process [36]. If the discriminator is unable to distinguish whether a feature originates from the source or target domain, it indicates that the model has learned domain-invariant features. Domain-adversarial neural networks [37] can be regarded as a prototype of DA approaches. Li et al. [38] proposed a deep domain FSL (DCFSL) method, which utilizes a domain-adversarial adaptive network to address the challenge of significant domain differences. Liu et al. [27] proposed a spectral shift mitigation (SSM) method that addresses cross-scene spectral variation via amplitude normalization and adjacency correction. In addition, Zhang et al. [39] introduced a single-source domain expansion network (SDEnet), which generates domain-extended samples using spatial–spectral randomization and contrastive adversarial training to enhance generalization under domain shift. Although the above methods show promise in addressing cross-domain tasks, existing domain-adversarial approaches focus on global alignment to reduce domain shift, often overlooking the importance of class-level alignment between source and target domains. Moreover, due to the larger amount of source-domain data, the model tends to learn the feature distribution of the source domain. In the absence of class-level alignment, it becomes difficult to extract class-level shared features between source and target domains. As a result, target-domain samples with class-level misalignment are incorrectly mapped near class boundaries in the feature space, which increases the number of edge points and further degrades classification performance, as illustrated in Fig. 1(b).

Moreover, in practical cross-domain classification tasks, the scarcity of labeled target-domain samples further impairs the performance of cross-domain HSI classification, thereby leading to a cross-domain few-shot learning scenario. To address this issue, researchers have adopted metric-based meta-learning methods [41]. The core idea of these methods is to construct a transferable feature metric space through episodic training, where the similarity measurement between query and support samples is utilized to tackle the challenge of limited labeled samples [42]. To enhance the reliability and discriminability of prototypes in few-shot learning, category-specific prototype self-refinement contrastive learning [43], mutual information enhancement FSL [44], and uncertainty-informed prototype contrastive learning [45] introduce self-refinement mechanisms, mutual information guidance, and uncertainty-informed contrastive objectives, respectively. However, in cross-domain few-shot classification tasks, the limited number of labeled samples in the target domain makes it challenging for the model to fully capture the complete feature distribution of each class. This problem is particularly evident in classes with high intraclass diversity, where the model tends to learn only the dominant feature patterns while failing to represent potential subdistributions. As a result, same-class samples with low feature similarity deviate from the main distribution in the feature space, residing in the tail regions and becoming edge points, which collectively lead to the long-tailed intraclass distribution, as shown in Fig. 1(c). However, existing prototype-based metric learning methods typically assume that intraclass features follow a uniform distribution and represent each class by averaging all its features. This approach struggles to model the above complex long-tailed distributions, leading to biased prototype estimation and impairing the model's classification performance [46], [47].

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In summary, existing methods for cross-domain few-shot HSI classification continue to face significant challenges. Due to limited feature extraction capability, class-level misalignment, and the scarcity of labeled samples in the target domain, a large number of edge points frequently emerge in the feature space. As shown in Fig. 1, these edge points typically lie far from their class centers and are easily confused with samples from nearby classes, thereby significantly degrading classification performance in the target domain.

To mitigate the adverse impact of edge points on cross-domain few-shot HSI classification, we propose the multi-Gaussian prototype metric with dual optimization (MGPDO). Specifically, to extract consistent features, we introduce a multiscale feature extraction (MSFE) module that captures both fine-grained and global features, thereby reducing the feature discrepancies of the same-class samples with diverse structures. To achieve class-level alignment under domain shift, we propose a cyclic domain triplet loss (CDTL) module that combines domain-adversarial networks with prototype contrastive learning. This module enhances the similarity between positive prototype pairs and suppresses the correlation among negative pairs, thereby improving class-level alignment across domains. To alleviate long-tailed distributions from limited target labeled samples, the multi-Gaussian prototype metric (MGPM) is designed to model intraclass distributions leveraging multiple Gaussian prototypes for each class, reducing class center estimation bias and accurately modeling the complex long-tailed feature distribution. The main contributions of this article are as follows.

1) A MSFE module with a spatial–spectral dual-branch architecture is proposed to enhance the extraction of class-consistent features by addressing spatial structure variation and spectral shift across domains, where the spatial branch captures local and global structures via multiscale convolutions and pyramid pooling, the spectral branch models interband correlations to preserve spectral continuity, and residual connections in both branches help retain low-level features and stabilize training.
2) A prototype-based CDTL module is introduced to achieve class-level alignment across domains. This module integrates a domain-adversarial network with cross-domain prototype contrastive learning, enhancing similarity between positive prototype pairs and suppressing the correlation between negative prototype pairs, thereby alleviating misclassification of edge points caused by domain misalignment.
3) To model the complex long-tailed distribution under limited labeled data, a meta-learning framework based on an MGPM is designed. By employing multiple Gaussian prototypes for each class, the framework effectively captures intraclass diversity, thereby improving the classification accuracy for tail-region edge points.

The rest of this article is organized as follows. Section II introduces the specific parts of MGPDO. The experimental results and analysis are presented in Section III. Finally, we conclude the article in Section IV.

Algorithm 1 Training Process of the MGPDO Framework

Input: S_s, Q_s, S_t, Q_t, and the number of training episodes N_{ep}.
Output: The total loss L_{total} of the model.
for episode = 1 to N_{ep} **do**
    Construct episodic batch from support and query sets
    Extract features using MSFE
    Calculate L_d by Eq. (1)
    Calculate prototypes $P_s^{c_s}, P_t^{c_t}$ by Eq. (2)
    Calculate $L^s_{\text{triplet}}, L^t_{\text{triplet}}$ by Eqs. (4), (5)
    $L_{\text{domain}} = L_d + \lambda(L^s_{\text{triplet}} + L^t_{\text{triplet}})$
    Calculate $L^s_{\text{cls}}, L^t_{\text{cls}}$ by Eqs. (17), (18)
    $L_{\text{total}} = L^s_{\text{cls}} + L^t_{\text{cls}} + L_{\text{domain}}$
    Backpropagate and update parameters
end for

II. Methodology

The proposed MGPDO framework aims to mitigate the issue of edge points in the feature space, resulting from the limited modeling capacity for structurally complex patches, misalignment of class-level features across domains, and the scarcity of labeled samples in the target domain, thereby enhancing the accuracy of cross-domain few-shot HSI classification. As shown in Fig. 2, the MGPDO framework consists of three main components. Specifically, it comprises the MSFE module for hyperspectral feature extraction, the CDTL module for cross-domain class-level alignment, and the MGPM module for modeling intraclass feature distributions. The overall process of the framework is divided into three stages. First, fixed-size HSI patches from the source and target domains are fed into the MSFE module. This module employs a multiscale spatial feature extraction mechanism with residual enhancement to capture fine-grained local details and global contextual information. Next, the extracted features are passed to the CDTL module, which introduces prototype-based contrastive learning combined with a cyclic optimization strategy. This facilitates effective cross-domain class alignment by increasing the similarity of positive prototype pairs and reducing the correlation among negative prototypes. Finally, the MGPM module models complex intraclass feature distributions by mapping each class to multiple Gaussian prototypes. This enables more accurate characterization of subtle intraclass variations, leading to improved classification accuracy. For a detailed description of the training process, refer to Algorithm 1.

As shown in Fig. 3, during the inference phase, the classification process for unlabeled samples begins with reducing the input dimensionality through the mapping layers. Subsequently, spatial–spectral embedded features are extracted using the trained MSFE module. Finally, the MGPM module performs classification on the unlabeled samples and produces the final classification maps.

A. MSFE Module

To effectively extract class-consistent features of patches with complex internal structures in HSIs and reduce the feature

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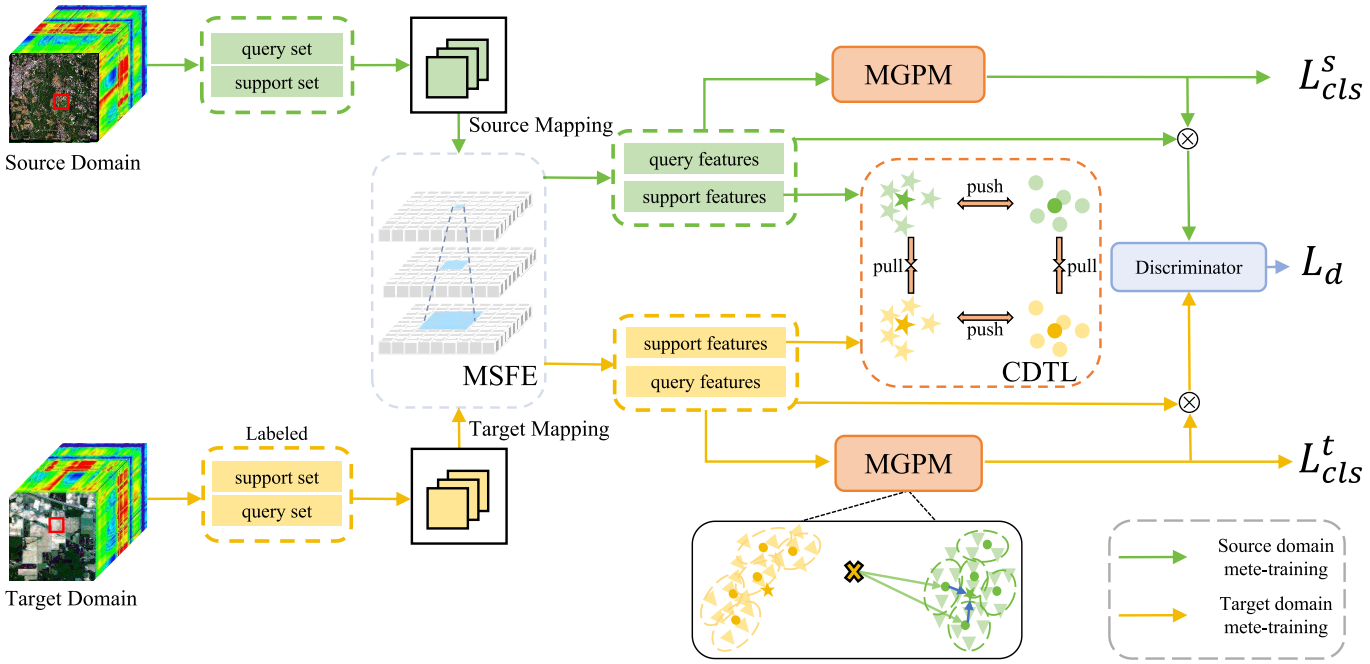


Fig. 2. Flowchart of the training phase in the proposed MGPDO framework, which integrates three core modules: the MSFE module, the CDTL module, and the MGPM module. MSFE captures both local spatial details and global semantic context via multiscale receptive fields and residual connections. CDTL performs class-level alignment using cyclic triplet loss based on source and target-domain prototypes. MGPM models intraclass diversity through Gaussian prototype distributions for robust metric learning.

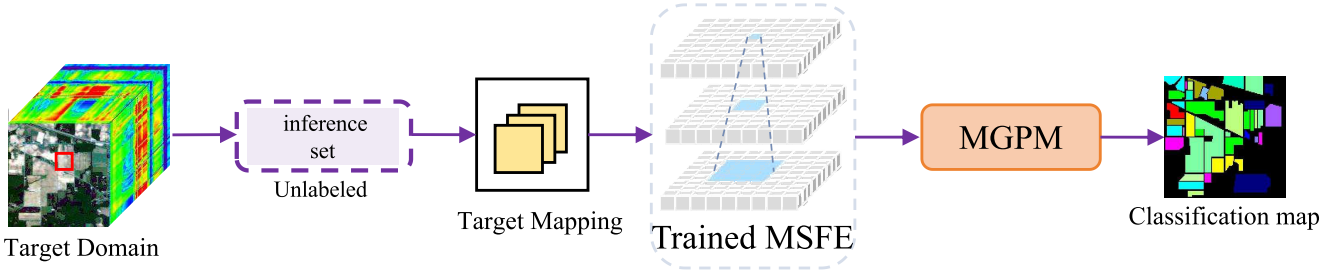


Fig. 3. Flowchart of the inference phase. Unlabeled target-domain samples are first passed through the trained MSFE module for spatial–spectral feature extraction. The extracted features are then classified using the MGPM module, which assigns predictions based on distances to the learned multi-Gaussian class prototypes.

discrepancies among same-class samples exhibiting structural diversity, we propose an MSFE module. As illustrated in Fig. 2, the proposed framework includes two mapping layers to align the dimensions of input data from the source and target domains before they are fed into the embedded multiscale feature extractor. Each mapping layer is implemented as a 2-D convolutional layer with d kernels of size $1 \times 1 \times$ ch, where ch and d denote the number of input and output channels, respectively. This MSFE module integrates three submodules: multiscale spatial feature extraction, spectral feature extraction, and spatial–spectral fusion, which operate in a collaborative manner. It leverages multiscale spatial information and spectral channel characteristics to extract highly discriminative features. The overall architecture of the MSFE module is illustrated in Fig. 4. First, the module effectively extracts key spatial and spectral information through a multiscale design combined with residual connections. This design captures both fine local and global information, which helps reduce domain shifts caused by differences in spatial resolution or image details due to different sensors or acquisition conditions. Meanwhile, the residual connections preserve low-level features and enhance the stability of gradient propagation. Finally, spatial and spectral features are fused to improve interclass separability. This fusion balances the complementary characteristics of spectral and spatial information, enabling the model to learn domain-invariant representations across different scenes and sensors. The design and function of each submodule are described as follows.

1) *Multiscale Spatial Feature Extraction Module:* To address the problem of boundary samples being susceptible to influence by other classes, we propose a multiscale spatial feature extraction module to capture the neighborhood information of target pixels in HSIs. The module compresses the spectral channels to 24

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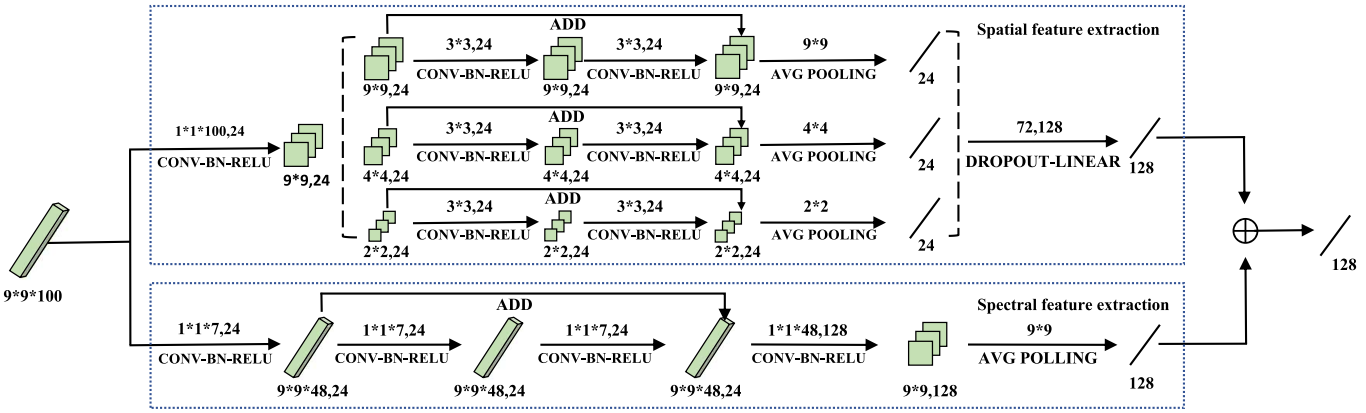


Fig. 4. Architecture of the proposed feature extractor. The feature extractor comprises three main modules: the multiscale spatial feature extraction module, the spectral feature extraction module, and the spatial–spectral feature fusion module. These modules collaboratively work to extract discriminative features from the input HSI. First, spatial features are extracted at multiple scales to capture spatial information across various resolutions. Simultaneously, spectral features are extracted to capture interchannel characteristics. Second, both spatial and spectral features are mapped into an embedding space of the same dimension to ensure consistency. Finally, the spatial and spectral features are fused with equal weights to generate the spatial–spectral joint features, which are then used for classification.

using a convolutional layer with 24, $1 \times 1 \times 100$ kernels and a stride of 1, while retaining spatial information. Then, three branches process spatial resolutions of 9×9, 4×4, and 2×2 (the latter is downsampled using max pooling), each containing a residual block [2, $1 \times 3 \times 3$ convolutional layers, 24 channels, with batch normalization (BN) and rectified linear unit (ReLU)]. The features are fused into a 72-D vector using global average pooling and then passed through a fully connected layer (72–128, with a dropout layer). The multiscale design enhances boundary sample separability by leveraging local and global features and improves cross-domain classification robustness through cross-scale feature consistency.

2) *Spectral Feature Extraction Module:* The spectral feature extraction module aims to extract the spectral information of target pixels from HSIs to capture the interchannel characteristics. This module consists of two convolutional layers and one residual block. The initial convolutional layer uses 24, $1 \times 1 \times 7$ kernels with a stride of (1, 1, 2) to compress redundant spectral features and reduce channel dimensions. The residual block contains two consecutive convolutional layers, each using 24, $1 \times 1 \times 7$ kernels with a stride of (1, 1, 1), followed by BN and ReLU activation to enhance the attention on important spectral channels and alleviate network degradation. Then, the second convolutional layer employs 128, $1 \times 1 \times 48$ kernels with a stride of (1, 1, 1) to perform feature transformation and dimensionality reduction, aligning with the spatial feature dimension. Finally, global average pooling with a 7×7 kernel generates a 128-D spectral feature vector.

3) *Spatial–Spectral Fusion Module:* The spatial–spectral fusion module combines spatial and spectral features through equal-weight averaging to generate joint features. This approach balances multiscale spatial context and spectral channel characteristics, enhancing feature discriminability, expanding interclass differences, and optimizing boundary sample separation and cross-domain classification performance.

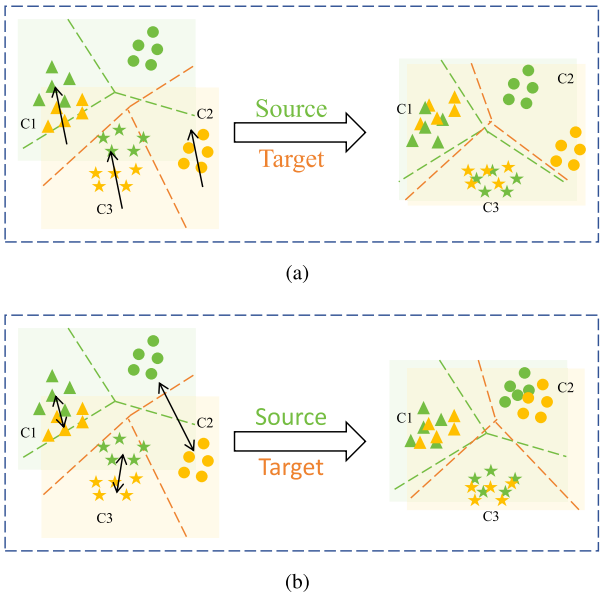


Fig. 5. Class-level domain distribution alignment is performed using (a) conditional domain-adversarial loss and (b) CDTL.

B. CDTL Module

To address the limitations of conventional conditional domain-adversarial networks (CDANs) in class-level alignment, we propose a CDTL within the MGPDO framework. Traditional CDANs primarily focus on global domain alignment through min–max adversarial training. While this strategy aligns the marginal distributions of source and target domains, it often fails to address the varying degrees of conditional distribution discrepancies across different classes between the two domains. As shown in Fig. 5, although features of classes C_1 and C_3 may be roughly aligned, class C_2

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exhibits clear misalignment, which leads to inaccurate feature mapping in the target domain, thereby forming edge points and degrading classification performance. To address the class-level distribution discrepancy across domains, we propose a CDTL. It constructs positive prototype pairs from the same class across domains and negative pairs from different classes within each domain, thereby promoting class-level alignment, reducing negative transfer, and improving the model's classification performance on the target domain.

The framework adopts a CDAN to perform domain-level alignment. Specifically, the feature extractor $\mathbf{f}$ maps each HSI patch $\mathbf{x} \in \mathbb{R}^{H\times W\times B}$ into a d-dimensional feature space, i.e., $\mathbf{f}(\mathbf{x}) \in \mathbb{R}^d$. The domain discriminator D is trained to distinguish whether $\mathbf{f}(\mathbf{x})$ originates from source or target-domain features. The adversarial loss is defined as

$$L_d = -\mathbb{E}_{\mathbf{x}_s\sim S}\left[\log D\left(\mathbf{f}\left(\mathbf{x}_s\right)\right)\right] - \mathbb{E}_{\mathbf{x}_t\sim T}\left[\log\left(1 - D\left(\mathbf{f}\left(\mathbf{x}_t\right)\right)\right)\right] \quad (1)$$

where $\mathbb{E}[\cdot]$ denotes the expectation over the corresponding data distribution, $\mathbf{x}_s$ and $\mathbf{x}_t$ are samples from the source and target domains, respectively. $D(\cdot)$ is the probability that the discriminator predicts that $\mathbf{x}$ is the source-domain sample, while $1 - D(\cdot)$ is the probability that the discriminator predicts that $\mathbf{x}$ is the target-domain sample. Discriminator D minimizes the above function, while feature extractor $\mathbf{f}$ maximizes it.

This adversarial loss encourages the feature extractor to generate indistinguishable features across domains, promoting global domain-level alignment. However, it lacks mechanisms to explicitly enforce class-level alignment, particularly in cases where different domain shifts exist in the conditional distributions of different classes. To overcome this, we introduce a prototype-based CDTL, which aligns class-level distributions via class prototypes and contrastive relationships. For each class c_s in the source domain and c_t in the target domain, we compute the corresponding class prototypes as the mean feature vectors of the support samples

$$\mathbf{p}_s^{c_s} = \frac{1}{K}\sum_{i=1}^{K}\mathbf{f}\left(\mathbf{x}_{s,i}^{c_s}\right), \quad \mathbf{p}_t^{c_t} = \frac{1}{K}\sum_{i=1}^{K}\mathbf{f}\left(\mathbf{x}_{t,i}^{c_t}\right) \quad (2)$$

where K is the number of support samples per class, and $\mathbf{x}_{s,i}^{c_s}$ and $\mathbf{x}_{t,i}^{c_t}$ represent support samples from class c_s in the source and class c_t in the target domains, respectively.

Next, we compute the Euclidean distance between each target prototype and all source prototypes

$$d\left(\mathbf{p}_t^{c_t}, \mathbf{p}_s^{c_s}\right) = \left\|\mathbf{p}_t^{c_t} - \mathbf{p}_s^{c_s}\right\|_2. \quad (3)$$

For each target-domain prototype, the closest source prototype in the embedding space is selected as the positive sample, as semantically similar classes typically exhibit closer distributions in the shared space. All other target prototypes are treated as negative samples. Based on this structure, we define a cyclic triplet loss, where the source prototype is used as the anchor

$$L_{\text{triplet}}^{s} = \max\left(d\left(\mathbf{p}_s^{c_s}, \mathbf{p}_t^{c_t}\right) - d\left(\mathbf{p}_s^{c_s}, \mathbf{p}_s^{c_s'}\right) + m, 0\right) \quad (4)$$

where $c_s' \neq c_s$, and m is a margin hyperparameter (set to 0.5) that enforces a sufficient distance gap between the positive and negative pairs.

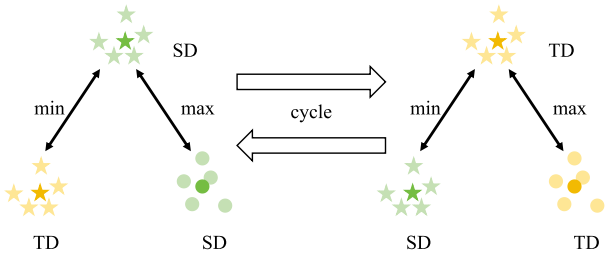


Fig. 6. Principle of cycle domain triplet loss mechanism.

To ensure symmetric optimization and mitigate alignment bias, we alternate domains and define the target prototype as the anchor

$$L_{\text{triplet}}^{t} = \max\left(d\left(\mathbf{p}_t^{c_t}, \mathbf{p}_s^{c_s}\right) - d\left(\mathbf{p}_t^{c_t}, \mathbf{p}_t^{c_t'}\right) + m, 0\right) \quad (5)$$

where $c_t' \neq c_t$, and the positive and negative pairs are similarly defined.

The total CDTL is calculated by summing over all source and target class triplets

$$L_{\text{cycle}} = \sum_{c_s} L_{\text{triplet}}^{s} + \sum_{c_t} L_{\text{triplet}}^{t}. \quad (6)$$

Finally, the overall DA objective integrates both global domain alignment and class-level alignment

$$L_{\text{domain}} = L_d + \lambda L_{\text{cycle}} \quad (7)$$

where λ is a hyperparameter that controls the contributions of the two loss terms.

As illustrated in Fig. 6, where the source domain is in green, the target domain in orange, and different shapes represent different classes, this cyclic alignment mechanism achieves effective class-level alignment across domains, while enlarging the margins between different classes within the same domain, thereby improving feature separability and effectively mitigating the problem of negative transfer. Compared with traditional CDAN-based strategies, our method achieves more robust DA by simultaneously addressing both the global and the class-conditional distribution shifts. The integration of CDAN and CDTL allows MGPDO to simultaneously address global and class-level domain shifts. While CDAN ensures global alignment conditioned on class predictions, CDTL enhances fine-grained class alignment via dual-anchor triplet losses, promoting intraclass compactness and interclass separability.

C. MGPM Module

In few-shot learning scenarios, the limited number of labeled samples often fails to capture the full diversity of intraclass features. Consequently, low-similarity samples within the same class may lie in the tail regions of the feature space, forming so-called edge points. These edge points pose significant challenges for conventional prototype-based metric learning methods, which represent each class using a single prototype (typically the mean of support features). This assumption of a unimodal and symmetric intraclass distribution is often invalid in real-world, cross-domain settings. To overcome this limitation, we propose the MGPM. Inspired by

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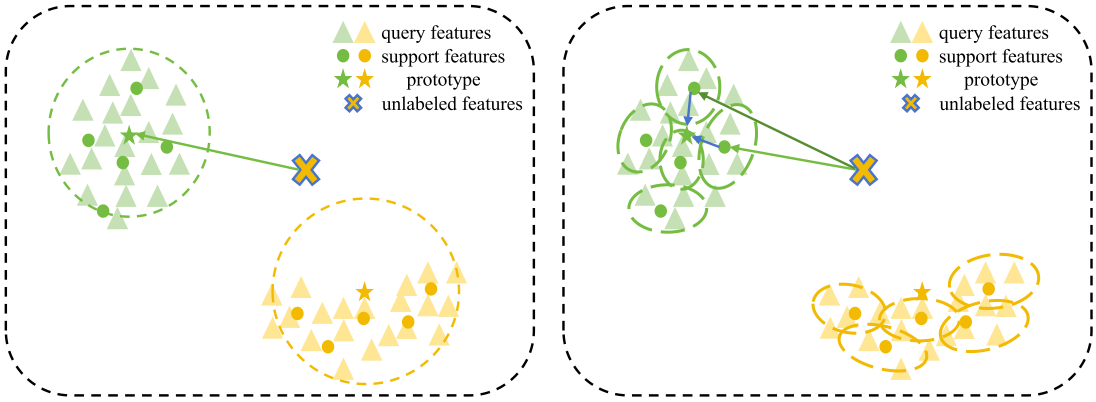


Fig. 7. Schematic of different metric methods. (a) Prototype metric. (b) MGPM.

recent advances in open-set recognition [48], MGPM models each class with multiple Gaussian prototypes, each defined by a learnable mean vector and covariance matrix. This one-to-many mapping enhances the model's capacity to capture long-tailed, submodal, and anisotropic feature distributions. As shown in Fig. 7, compared with existing methods that overlook intraclass diversity [49], MGPM offers a more flexible and robust representation of class-level variation, effectively addressing the challenges posed by edge points.

In few-shot learning, the scarcity of support samples necessitates fully leveraging each sample. For class c, we map its K support samples $\mathbf{x}_{c,i}$ to the feature space using a feature extractor $\mathbf{f}(\cdot)$, and define each as a Gaussian prototype for that class

$$\boldsymbol{\mu}_{c,i} = \mathbf{f}\left(\mathbf{x}_{c,i}\right). \tag{8}$$

Each Gaussian prototype $\boldsymbol{\mu}_{c,i}$ is associated with a covariance matrix $\boldsymbol{\Sigma}_{c,i} \in \mathbb{R}^{d\times d}$, which characterizes the subdistribution of class c in the feature space. The covariance matrix is computed based on all support samples of the class

$$\boldsymbol{\Sigma}_{c,i} = \frac{1}{K-1}\sum_{j=1}^{K}\left(\mathbf{f}\left(\mathbf{x}_{c,j}\right)-\boldsymbol{\mu}_{c,i}\right)\left(\mathbf{f}\left(\mathbf{x}_{c,j}\right)-\boldsymbol{\mu}_{c,i}\right)^{\mathrm{T}}. \tag{9}$$

To ensure numerical stability in few-shot tasks, the covariance matrix is regularized by the Sigmoid

$$\boldsymbol{\Sigma}'_{c,i} = \mathrm{Sigmoid}\left(\boldsymbol{\Sigma}_{c,i}\right). \tag{10}$$

For a query sample $\mathbf{x}$, its distance to each of the K Gaussian prototypes in class c is computed as

$$s_{c,i} = \exp\left(-\frac{1}{2}\left(\mathbf{f}\left(\mathbf{x}\right)-\boldsymbol{\mu}_{c,i}\right)^{\mathrm{T}}\left(\boldsymbol{\Sigma}'_{c,i}\right)^{-1}\left(\mathbf{f}\left(\mathbf{x}\right)-\boldsymbol{\mu}_{c,i}\right)\right) \tag{11}$$

where $s_{c,i}$ measures the distance between the query sample $\mathbf{x}$ and the Gaussian prototype $\boldsymbol{\mu}_{c,i}$, reflecting cross-dimensional correlations as defined by $\boldsymbol{\Sigma}'_{c,i}$. To aggregate the information from all K Gaussian prototypes, we introduce a weighting mechanism and classify via a sparsely connected multilayer perceptron (MLP). Specifically, the class center is computed as the mean vector of its corresponding Gaussian prototypes

$$\mathbf{v}_c = \frac{1}{K}\sum_{i=1}^{K}\boldsymbol{\mu}_{c,i} \in \mathbb{R}^d. \tag{12}$$

Next, the Euclidean distance between each prototype $\boldsymbol{\mu}_{c,i}$ and the class center $\mathbf{v}_c$ is computed as

$$d_{c,i} = \left\|\boldsymbol{\mu}_{c,i}-\mathbf{v}_c\right\|_2. \tag{13}$$

Then, a softmax function is applied over the prototype-to-center distances to assign higher weights to prototypes that are closer to the class center

$$w_{c,i} = \frac{\exp\left(-d_{c,i}\right)}{\sum_{j=1}^{K}\exp\left(-d_{c,j}\right)}. \tag{14}$$

The weighted Gaussian distances $s^w_{c,i}$ are computed as

$$s^w_{c,i} = w_{c,i}\cdot s_{c,i}. \tag{15}$$

Finally, these weighted distances are sorted and concatenated to form a K-dimensional vector $\mathbf{s}^w_c \in \mathbb{R}^K$. The vectors for all C classes are concatenated and fed into a sparsely connected MLP, which outputs a probability vector $\mathbf{P} \in \mathbb{R}^C$, representing the predicted probabilities for each class. The final label is assigned to the class with the highest probability

$$\hat{y} = \arg\max\left(\mathrm{Softmax}\left(\mathrm{MLP}\left(\left[\mathbf{s}^w_1,\ldots,\mathbf{s}^w_C\right]\right)\right)\right) \tag{16}$$

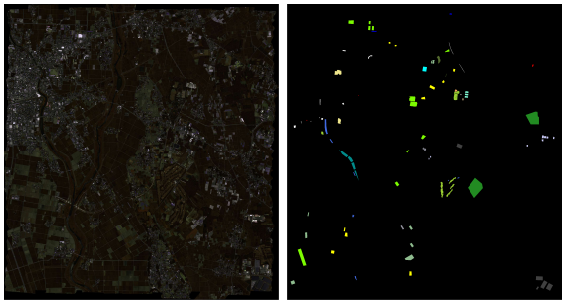
where $\hat{y} \in \{1,\ldots,C\}$ is the predicted class label. The classification loss function on the source-domain episode is defined as

$$L^s_{\mathrm{cls}} = \mathbb{E}_{Q_s}\left[-\sum_{(\mathbf{x},y)\in Q_s}\log p\left(y=k\mid\mathbf{x}\right)\right] \tag{17}$$

where Q_s is the query set from the source domain and $p(y = k \mid \mathbf{x})$ is the predicted probability for class k. Similarly, the classification loss for the target-domain episode can be defined as

$$L^t_{\mathrm{cls}} = \mathbb{E}_{Q_t}\left[-\sum_{(\mathbf{x},y)\in Q_t}\log p\left(y=k\mid\mathbf{x}\right)\right] \tag{18}$$

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(a) (b)

Fig. 8. Pseudo-color image and ground-truth map of CH. (a) Pseudo-color image. (b) Ground-truth map of the training set.

where Q_t is the query set from the target domain. During the alternating training of source and target episodes, samples from the source and target domains are separately used to compute classification losses, helping the model adapt to data from both domains. The overall classification loss is defined as

$$L_{\text{cls}} = L_{\text{cls}}^{s} + L_{\text{cls}}^{t}. \tag{19}$$

Finally, the total loss function of the method is defined as

$$L_{\text{total}} = L_{\text{domain}} + L_{\text{cls}} \tag{20}$$

where L_{domain} denotes the domain alignment loss that minimizes the discrepancy between the source and target domains, and L_{cls} represents the classification loss used to optimize feature discriminability. This combination ensures that the model learns both domain-invariant and class-discriminative features for effective cross-domain few-shot learning.

III. Experimental Results and Analysis

A. Dataset Description

We use four publicly available HSI datasets, including Chikusei (CH), Indian Pines (IP), Salinas (SA), and the University of Pavia (PU), to evaluate the effectiveness of the proposed method. Following the FSL experimental settings, CH is used as the source-domain dataset, and the other three are used as the target-domain datasets.

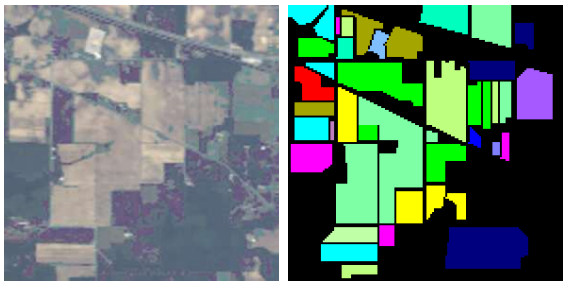
1) CH Dataset: The CH dataset was taken by the Headwall Hyperspec-VNIR-C sensor in 2014 at CH, Tsukushi, Japan. The dataset contains 128 wavelengths in the band range of 340–1018 nm, with a spatial size of 2517 × 2335 pixels and a spatial resolution of 2.5 m. The dataset contains a total of 19 different classes, covering both urban and rural areas. The details of this dataset are shown in Table I, and the false-color image and ground truth are shown in Fig. 8.

2) IP Dataset: The IP dataset was taken by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in Indiana. With 224 bands from 400 to 2500 nm, 200 bands were retained after removing the spectra covering the water absorption range. The image contains 145 × 145 pixels with a spatial resolution of 20 m/pixel. The data has a total of 16 crop classes. The data contains 16 crop classes. The details of the IPs dataset are shown in Table II, and the false-color image and ground truth are shown in Fig. 9.

TABLE I

Class Name and Sample Size for CH Dataset

Class	Name	Samples
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
	Total	77592



(a) (b)

Fig. 9. Pseudo-color image and ground-truth map of IP. (a) Pseudo-color image. (b) Ground-truth map of the training set.

TABLE II

Class Name and Sample Size for the IP Dataset

Class	Name	Samples
1	Alfalfa	46
2	Corn-notil	1428
3	Corn-mintill	830
4	Corn	237
5	Grass pasture	483
6	Grass trees	730
7	Grass pasture mowed	28
8	Hay windrowed	478
9	Oats	20
10	Soybean notill	972
11	Soybean mintill	2455
12	Soybean clean	593
13	Wheat	205
14	Woods	1265
15	Buildings Grass Trees Drives	386
16	Stone Steel Towers	93
	Total	10249

3) SA Dataset: The SA dataset was captured by the AVIRIS sensor in the SA Valley, CA, USA. The spatial resolution of the data is 3.7 m/pixel, and the size is 512 × 217. The original data has 224 bands, and after removing the bands with serious water vapor absorption, 204 bands remain. The data contains 16 crop classes. The details of the SA dataset are shown in

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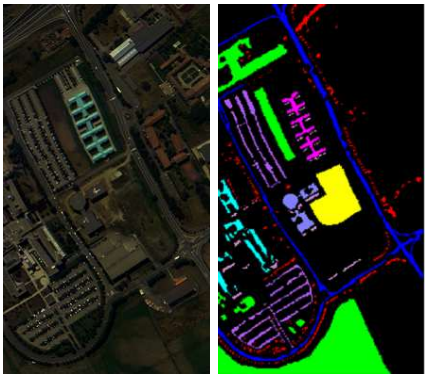


(a) (b)

Fig. 10. Pseudo-color image and ground-truth map of SA. (a) Pseudo-color image. (b) Ground-truth map of the training set.

TABLE III
CLASS NAME AND SAMPLE SIZE FOR THE SA DATASET

Class	Name	Samples
1	Brocoli green weeds 1	2009
2	Brocoli green weeds 2	3726
3	Fallow	1976
4	Fallow rough plow	1394
5	Fallow smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes untrained	11271
9	Soil vinyard develop	6203
10	Corn senesced green weeds	3278
11	Lettuce romaine 4wk	1068
12	Lettuce romaine 5wk	1927
13	Lettuce romaine 6wk	916
14	Lettuce romaine 7wk	1070
15	Vinyard untrained	7268
16	Vinyard untrained trellis	1807
	Total	54129



(a) (b)

Fig. 11. Pseudo-color image and ground-truth map of PU. (a) Pseudo-color image. (b) Ground-truth map of the training set.

Table III, and the false-color image and ground-truth image are shown in Fig. 10.

4) PU Dataset: The PU dataset was acquired by the reflection optical system imaging Spectrometer (ROSIS) sensor in the city of Pavia, Italy, in 2003. The PU dataset contains nine classes. The dataset contains 103 usable spectral bands after removing 12 noise bands; the spatial size of the dataset is 610×340, while the spatial resolution is 1.3 m. The details of the PU dataset are shown in Table IV, and the false-color image and ground truth are shown in Fig. 11.

TABLE IV
CLASS NAME AND SAMPLE SIZE FOR THE PU DATASET

Class	Name	Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-blocking	3682
9	Shadows	947
	Total	42776

B. Experimental Setups

The deep learning framework used is PyTorch version 1.7.1, and dual GeForce RTX 3090 Ti GPUs are employed for the hardware setup. In the MGPDO framework, input samples are constructed as patches of dimensions $9 \times 9 \times B$, where B represents the number of spectral bands. The model is trained for 10000 iterations, which is empirically sufficient for network convergence. The MGPDO is trained via the Adam optimizer, and the learning rate is set to 0.001. In the few-shot learning paradigm, we define C as the number of classes per episode and K as the number of support samples per class. Each episode corresponds to a K-shot, C-way classification task, where C is determined by the number of classes in the target dataset (e.g., nine for the PU dataset, and 16 for both the IP and SA datasets). The metatask settings follow this approach described in [38]. For each metatask, K is set to 5, which means that each class has five labeled samples as the support set. N_Q is set to 19, which means that 19 unlabeled samples of each class are taken as the query set. The classification performance of all methods is evaluated using several metrics, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (KC).

C. Parameter Tuning

1) Effect of the CDTL Weight Coefficient λ: In this section, λ serves as the coefficient for the CDTL, regulating the balance between domain alignment and class alignment. To analyze the impact of λ on model performance, we conducted parameter sensitivity experiments across multiple target domains, selecting different values from the set $\{1e-3, 1e-2, 1e-1, 1e+0, 1e+1\}$ for evaluation.

λ is a crucial hyperparameter for class alignment optimization. When λ is too low, the optimization primarily focuses on aligning the global feature distribution, effectively reducing the overall distribution discrepancy between the source and target domains. However, due to insufficient class-level constraints, target-domain features may still suffer from class confusion, resulting in ambiguous classification decision boundaries and

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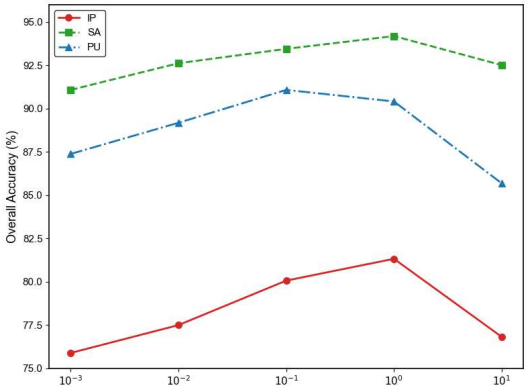


Fig. 12. OA (%) performance with different λ values on the three datasets.

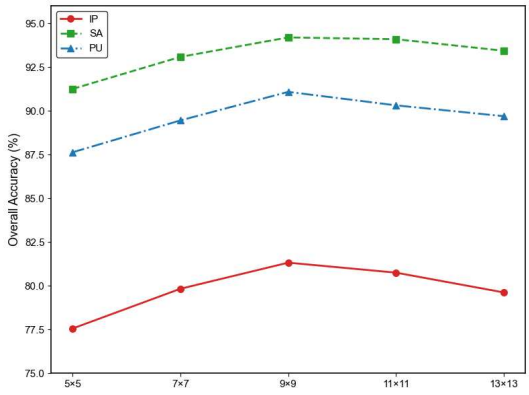


Fig. 13. OA (%) performance with different patch sizes on the three datasets.

TABLE V
OA (%) PERFORMANCE WITH DIFFERENT M ON THREE DATASETS

Dataset	$m = 0.1$	$m = 0.3$	$m = 0.5$	$m = 0.7$	$m = 0.9$
IP	79.85	80.63	**81.32**	80.97	80.21
SA	92.76	93.68	**94.19**	93.85	93.02
PU	89.75	90.56	91.08	**91.23**	90.27

TABLE VI
OA (%) PERFORM WITH DIFFERENT REGULARIZATION FUNCTIONS IN THE MGPM MODULE

Dataset	Sigmoid	Tanh	Softplus	ReLU	LeakyReLU
IP	**81.32**	79.72	78.98	78.49	79.61
SA	94.19	**94.42**	87.91	91.84	90.04
PU	**91.08**	89.37	89.11	89.32	90.17

degraded performance. Conversely, when λ is too high, class-level alignment dominates, forcing target-domain features to excessively align with source-domain class prototypes. Since the class sets of the source and target domains may not be entirely consistent, excessive alignment can mislead certain target-domain classes toward incorrect class prototypes, ultimately weakening the model's generalization ability.

To ensure a balanced optimization between domain and class alignment, we conducted experiments to determine the optimal value of λ for different datasets. As shown in Fig. 12, the results indicate that the optimal λ value for the IP and SA datasets is $1e+0$, while for the PU dataset, the optimal value is $1e-1$. These findings highlight the importance of selecting an optimal λ to improve cross-domain classification performance across different datasets.

2) Effect of the m in CDTL: The margin parameter m in the CDTL governs the separation between positive and negative pairs in the embedding space. To assess its impact, we conducted sensitivity experiments by varying m in $\{0.1, 0.3, 0.5, 0.7, 0.9\}$, and report the classification performance across all target datasets.

As shown in Table V, performance varies nonmonotonically with m. Specifically, for the IP dataset, the OA reaches its highest value of 81.32% when the margin is set to 0.5. Similarly, the SA dataset achieves the best performance of 94.19% under the same margin setting. For the PU dataset, although a slightly higher accuracy of 91.23% is observed when the margin is set to 0.7, the difference compared with 0.5 (91.08%) is marginal. These results indicate that setting the margin to 0.5 provides a robust tradeoff between enhancing interclass separation and maintaining intraclass compactness. Therefore, we adopt 0.5 as the default margin value in our framework.

3) Effect of Input Patch Size: To evaluate the effect of spatial scale in the MSFE module, we conduct a hyperparameter study by varying the size of the input patches. Since the 9×9, 4×4, and 2×2 branches in the MSFE module are implemented through progressive downsampling operations, the initial input patch size directly determines the resolution of multiscale representations. We experiment with five input patch sizes: 5×5, 7×7, 9×9, 11×11, and 13×13, while keeping other settings fixed.

The experimental results are presented in Fig. 13. It can be observed that the classification accuracy increases with the input patch size up to 9×9, after which the performance begins to decline. This trend suggests that a 9×9 spatial scale provides an optimal tradeoff between capturing sufficient contextual information and mitigating the influence of irrelevant or noisy surrounding pixels. These results empirically validate the selection of 9×9 as the base scale for progressive downsampling in the proposed MSFE module.

4) Effect of the Regularization Function in MGPM: To alleviate the numerical instability in covariance estimation caused by limited support samples in the MGPM module, we introduce a nonlinear regularization function, Sigmoid, to constrain the estimated values. Five commonly used activation functions, including Sigmoid, Tanh, Softplus, ReLU, and LeakyReLU, are empirically evaluated on three datasets.

As shown in Table VI, Sigmoid consistently delivers stable and high performance across all datasets. While Tanh slightly outperforms Sigmoid on SA, it performs notably worse on IP and PU. In contrast, Softplus and ReLU result in larger accuracy degradation, suggesting weaker regularization in the few-shot setting. Theoretically, Sigmoid maps values to a bounded range $(0, 1)$, which suppresses extreme estimates and mitigates the amplification of noisy correlations. This improves the stability of the Mahalanobis-like distance metric in MGPM. Based on these findings, we adopt Sigmoid as the default regularization function.

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D. Classification Performance of MGPDO and Comparison Methods

To evaluate the classification performance of the proposed MGPDO in cross-domain few-shot scenarios, we compare it with seven related methods, including the conventional deep learning methods, spectral-spatial residual network (SSRN), mixed use of 3-D and 2-D CNN (MCNN), and spectral partitioning residual network (SPRN); and the cross-domain few-shot learning methods, DCFSL, graph information aggregation cross-domain few-shot learning (Gia-CFSL), adaptive domain-adversarial FSL (ADAFSL), refined prototypical contrastive learning network for FSL (RPCL-FSL), and cross-modal alignment and supervised contrastive learning (CASCL).

1) SSRN [18] introduces an end-to-end spectral–spatial residual network for HSI classification, using raw 3-D cubes as input. The network employs spectral and spatial residual blocks to learn discriminative features.
2) MCNN [50] integrates 3-D and 2-D convolutions for spectral–spatial feature extraction, followed by covariance pooling to capture second-order feature statistics, enhancing discrimination capability.
3) SPRN [51] partitions spectral bands into nonoverlapping subbands and deploys parallel spectral–spatial residual blocks, followed by grouped convolution-based feature fusion for classification.
4) DCFSL [38] combines few-shot learning with DA to address domain shift in HSI classification. It uses a conditional adversarial DA strategy for domain alignment and performs FSL for both source and target classes, enabling knowledge transfer and improved classification.
5) Gia-CFSL [52] combines few-shot learning with domain alignment through graph information aggregation. It introduces an intradomain distribution extraction block (IDE-block) to capture nonlocal spatial relationships and a cross-domain similarity aware block (CSA-block) to align feature and distribution similarities, mitigating domain shift.
6) ADAFSL [53] employs an adaptive domain-adversarial strategy to align global and class-specific conditional distributions, assigning varying weights to adversarial losses based on classwise distribution discrepancies and learning domain-invariant features through an adaptive weighting mechanism.
7) RPCL-FSL [40] integrates supervised contrastive learning (SCL) with few-shot learning to stabilize prototypes and address domain shift in small-sample HSI classification. It imposes CL, self-calibration (SC), and cross-calibration (CC) constraints on prototypes, enhancing their stability.
8) CASCL [54] introduces a cross-domain few-shot HSI classification framework that combines cross-modal alignment (CA) and SCL. It leverages semantic guidance from text prototypes to enhance image prototype quality and applies neighborhood-based SCL to suppress noise and redundancy, improving feature discriminability in the target domain.

To ensure a fair comparison, the training and testing classes remain the same for the conventional deep learning methods, SSRN, MCNN, and SPRN. In the target domain, five labeled samples from each class are randomly selected as the training set, while the remaining samples are used as the testing set. For the cross-domain FSL methods DCFSL, Gia-CFSL, ADAFSL, RPCL-FSL, CASCL, and the proposed MGPDO method, five labeled samples from each class in the target domain are also randomly selected as training samples, with the remaining samples used for testing. Meanwhile, the CH dataset is used as the source domain, and 200 samples from each base class are randomly selected for training. Considering the effect of random sampling, all experiments were conducted ten times, and the average results were reported.

1) Quantitative Results: Tables VII–IX present the classification results of all methods on the IP, SA, and PU datasets with five labeled samples per class. Table VII demonstrates that the proposed MGPDO method significantly outperforms all other methods on the IP dataset. Specifically, in terms of the three evaluation dimensions of OA, AA, and KC, the cross-domain FSL methods RPCL-FSL and CASCL exhibit strong performance. Compared with the RPCL-FSL method, the proposed MGPDO achieves improvements of 6.21%, 3.49%, and 6.81% in OA, AA, and KC, and compared with the CASCL method, it improves by 1.59%, 1.19%, and 2.22%, respectively. Furthermore, the proposed MGPDO method achieves the highest classification accuracy in ten classes, including eight fine-grained categories with highly similar intraclass characteristics (e.g., corn, grass, and soybean), which account for the majority of challenging categories in the IP dataset. This result highlights the effectiveness of MGPDO in addressing fine-grained cross-domain classification.

Table VIII presents the classification performance of different methods on the SA dataset. Among all the compared methods, the conventional deep learning method, SPRN, and the cross-domain FSL method CASCL perform outstandingly compared with other methods. However, the proposed MGPDO method consistently outperforms all compared methods in terms of OA. Specifically, in terms of the three evaluation dimensions of OA, AA, and KC, the MGPDO method surpasses the best-performing conventional deep learning method, SPRN, by 4.14%, 2.88%, and 4.91%, respectively. Comparedwith the best-performing FSL method CASCL, the MGPDO further improves the OA, AA, and KC by 0.55%, 1.10%, and 0.47%, respectively.

Table IX presents the classification results of compared methods on the PU dataset, demonstrating that the proposed MGPDO method consistently outperforms all compared approaches. Specifically, in terms of the three evaluation metrics of OA, AA, and KC, the MGPDO method surpasses the best-performing cross-domain FSL method CASCL by 1.33%, 3.34%, and 2.98%, respectively, and outperforms the best-performing conventional deep learning method SPRN by 11.56%, 12.37%, and 14.86%. Moreover, the MGPDO method achieves the highest classification accuracy in the fourth class “trees” and sixth class “bare soil”, which are the most challenging to classify, further demonstrating the superiority of the proposed approach. Moreover, the standard deviations reported in Tables VII–IX further indicate that the

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TABLE VII

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), AND KC (%) ON IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SSRN	MCNN	SPRN	DCFSL	Gia-CFSL	ADAFSL	RPCL-FSL	CASCL	Ours
1	99.01 ± 0.25	93.24 ± 2.67	98.94 ± 1.56	97.80 ± 1.78	93.41 ± 2.63	**100.00**	98.54 ± 0.86	97.56 ± 1.86	**100.00**
2	42.91 ± 5.11	46.14 ± 4.08	54.76 ± 2.75	45.14 ± 1.27	47.98 ± 4.12	51.03 ± 2.86	62.58 ± 3.86	**73.11 ± 3.43**	71.98 ± 3.64
3	58.84 ± 8.26	46.24 ± 5.57	51.14 ± 6.48	46.88 ± 7.04	49.35 ± 3.17	64.10 ± 5.17	57.09 ± 6.54	62.85 ± 2.77	**63.56 ± 2.30**
4	81.49 ± 3.27	76.57 ± 2.95	83.78 ± 1.81	75.60 ± 2.83	76.51 ± 2.07	81.21 ± 3.12	93.23 ± 1.96	**95.32 ± 2.54**	90.37 ± 2.62
5	78.63 ± 2.07	66.39 ± 3.71	73.53 ± 4.99	67.74 ± 2.63	66.65 ± 3.92	79.75 ± 1.92	78.16 ± 1.48	77.86 ± 1.77	**83.47 ± 0.39**
6	95.75 ± 2.78	87.15 ± 2.97	90.91 ± 4.43	86.50 ± 3.07	84.14 ± 2.69	92.63 ± 2.69	90.06 ± 2.59	94.86 ± 2.87	**96.37 ± 1.92**
7	99.55 ± 0.22	97.27 ± 0.97	99.65 ± 0.74	99.57 ± 0.27	99.57 ± 0.57	**100.00**	**100.00**	**100.00**	**100.00**
8	76.91 ± 3.74	78.14 ± 1.57	84.13 ± 3.16	76.03 ± 1.68	83.76 ± 3.97	92.30 ± 4.97	90.51 ± 1.53	95.32 ± 2.48	**99.93 ± 1.12**
9	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
10	60.51 ± 5.12	65.32 ± 6.81	67.95 ± 5.63	57.08 ± 1.73	58.23 ± 4.68	59.54 ± 3.97	73.80 ± 2.59	**76.25 ± 4.87**	75.32 ± 2.77
11	51.14 ± 6.12	62.34 ± 4.57	65.22 ± 1.68	61.45 ± 5.74	61.32 ± 2.68	69.93 ± 3.97	69.56 ± 2.59	78.14 ± 5.83	**80.98 ± 3.53**
12	53.59 ± 11.93	53.17 ± 4.69	49.22 ± 8.32	48.69 ± 2.17	45.83 ± 5.07	59.93 ± 1.63	**64.40 ± 7.54**	63.20 ± 3.77	59.41 ± 6.21
13	99.29 ± 0.14	97.14 ± 2.19	97.58 ± 3.03	95.35 ± 2.68	96.45 ± 2.97	97.20 ± 1.91	98.50 ± 1.53	98.33 ± 1.48	**99.33 ± 0.47**
14	87.98 ± 4.12	87.24 ± 1.52	90.22 ± 1.23	89.48 ± 3.74	87.83 ± 3.68	87.86 ± 3.97	90.55 ± 1.91	89.13 ± 3.53	**92.67 ± 2.13**
15	75.21 ± 3.12	75.96 ± 2.62	80.18 ± 2.31	67.72 ± 4.73	70.21 ± 1.68	71.50 ± 2.67	80.63 ± 4.59	86.75 ± 3.87	**91.69 ± 3.79**
16	97.71 ± 1.74	97.58 ± 2.21	96.51 ± 1.95	96.82 ± 3.68	96.02 ± 2.97	97.73 ± 1.91	97.05 ± 1.53	**97.78 ± 2.48**	95.45 ± 1.71
OA(%)	65.72 ± 3.77	66.17 ± 3.21	67.32 ± 2.08	64.89 ± 3.50	65.44 ± 2.23	71.91 ± 3.15	75.11 ± 1.97	79.73 ± 1.48	**81.32 ± 1.62**
AA(%)	79.22 ± 1.62	80.32 ± 2.83	81.28 ± 1.12	75.75 ± 2.44	76.08 ± 1.48	81.54 ± 1.49	84.04 ± 1.10	86.34 ± 1.73	**87.53 ± 0.31**
KC(%)	61.72 ± 3.90	62.57 ± 3.67	63.32 ± 2.93	60.36 ± 3.96	61.02 ± 2.53	68.23 ± 3.39	71.94 ± 2.17	76.53 ± 1.86	**78.75 ± 1.73**

TABLE VIII

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), AND KC (%) ON SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SSRN	MCNN	SPRN	DCFSL	Gia-CFSL	ADAFSL	RPCL-FSL	CASCL	Ours
1	98.53 ± 0.86	98.41 ± 1.51	99.13 ± 0.69	99.45 ± 0.34	98.09 ± 0.63	99.19 ± 0.41	98.95 ± 0.89	99.45 ± 0.24	**100.00**
2	99.67 ± 0.13	99.76 ± 0.32	99.69 ± 0.33	98.82 ± 0.88	99.41 ± 0.43	99.68 ± 0.28	99.09 ± 0.77	99.62 ± 0.43	**100.00**
3	96.87 ± 2.23	91.96 ± 3.51	97.01 ± 1.83	97.31 ± 1.81	95.46 ± 3.22	97.10 ± 2.15	96.09 ± 3.21	**99.84 ± 0.15**	99.81 ± 0.26
4	99.31 ± 0.62	99.25 ± 0.52	99.36 ± 0.52	99.26 ± 0.61	98.52 ± 1.15	99.57 ± 0.83	99.09 ± 0.46	99.49 ± 0.31	**99.81 ± 0.20**
5	95.79 ± 1.45	92.70 ± 3.72	94.58 ± 1.45	91.52 ± 1.52	86.55 ± 2.63	95.74 ± 3.38	90.83 ± 1.55	95.94 ± 1.12	**97.27 ± 1.52**
6	99.66 ± 0.28	99.52 ± 0.42	99.47 ± 0.53	**99.84 ± 0.25**	99.25 ± 0.32	99.26 ± 0.42	98.76 ± 1.27	99.82 ± 0.35	99.75 ± 0.42
7	99.90 ± 0.03	98.58 ± 1.21	99.53 ± 0.32	99.69 ± 0.66	99.28 ± 0.32	99.30 ± 0.53	99.17 ± 0.49	**99.91 ± 0.29**	99.63 ± 0.13
8	66.17 ± 2.68	74.57 ± 3.92	78.74 ± 2.68	79.48 ± 1.54	81.31 ± 1.71	77.67 ± 1.45	82.24 ± 1.62	84.52 ± 2.51	**86.81 ± 1.81**
9	99.79 ± 0.27	99.59 ± 0.31	99.13 ± 0.37	99.32 ± 0.55	97.76 ± 1.08	99.66 ± 0.34	99.94 ± 0.05	99.71 ± 0.25	**100.00**
10	90.73 ± 2.85	83.42 ± 3.08	84.98 ± 2.85	86.78 ± 1.92	83.73 ± 2.78	84.64 ± 3.65	90.00 ± 2.83	92.69 ± 1.72	**93.39 ± 1.95**
11	97.34 ± 1.08	96.61 ± 3.14	96.99 ± 1.82	97.01 ± 2.07	93.56 ± 1.11	99.08 ± 0.76	99.13 ± 0.38	98.30 ± 1.57	**99.92 ± 0.09**
12	99.88 ± 0.19	98.93 ± 1.34	99.13 ± 0.39	99.17 ± 0.68	99.19 ± 0.32	**99.92 ± 0.15**	98.36 ± 0.81	99.81 ± 0.68	99.75 ± 0.26
13	99.01 ± 0.48	98.30 ± 1.65	98.34 ± 1.37	97.85 ± 1.45	94.27 ± 3.52	99.09 ± 0.59	97.09 ± 1.47	**99.66 ± 0.42**	99.58 ± 0.57
14	98.13 ± 1.34	98.24 ± 1.51	98.36 ± 0.93	98.43 ± 0.72	98.46 ± 1.38	99.44 ± 0.28	97.50 ± 0.35	98.24 ± 1.33	**99.64 ± 0.34**
15	81.47 ± 6.55	76.38 ± 4.82	80.54 ± 4.55	75.73 ± 6.62	74.51 ± 9.71	84.40 ± 8.48	84.59 ± 6.54	83.91 ± 5.54	**85.98 ± 9.62**
16	94.43 ± 2.85	92.22 ± 3.08	91.03 ± 1.85	92.75 ± 2.92	90.32 ± 2.78	92.62 ± 1.65	93.14 ± 1.72	94.23 ± 2.93	**94.23 ± 2.72**
OA(%)	89.13 ± 2.59	89.34 ± 2.19	90.05 ± 1.53	90.51 ± 0.99	89.75 ± 0.80	91.43 ± 0.47	92.33 ± 0.78	93.64 ± 0.63	**94.19 ± 0.94**
AA(%)	94.79 ± 0.55	94.04 ± 1.14	94.11 ± 0.96	94.52 ± 0.66	93.10 ± 1.19	95.28 ± 0.58	95.22 ± 0.71	95.89 ± 0.42	**96.99 ± 0.40**
KC(%)	87.95 ± 2.80	88.17 ± 2.41	88.63 ± 1.91	89.45 ± 1.09	88.60 ± 0.90	90.49 ± 0.52	91.47 ± 0.83	93.07 ± 0.69	**93.54 ± 1.05**

TABLE IX

CLASS-SPECIFIC CLASSIFICATION ACCURACY (%), OA (%), AA (%), AND KC (%) ON PU DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SSRN	MCNN	SPRN	DCFSL	Gia-CFSL	ADAFSL	RPCL-FSL	CASCL	Ours
1	91.21 ± 2.14	81.23 ± 1.71	82.63 ± 3.87	80.37 ± 1.89	88.12 ± 2.56	87.81 ± 2.63	87.57 ± 1.59	89.50 ± 3.48	**92.09 ± 1.09**
2	**93.94 ± 0.87**	86.31 ± 2.21	86.85 ± 1.96	85.49 ± 1.92	85.95 ± 2.22	86.92 ± 2.85	87.53 ± 1.91	86.44 ± 1.63	92.50 ± 0.78
3	55.30 ± 4.07	57.62 ± 3.89	58.88 ± 2.12	63.74 ± 3.54	66.44 ± 3.78	59.96 ± 4.21	67.62 ± 3.85	75.64 ± 3.69	**89.85 ± 3.21**
4	95.21 ± 0.92	94.23 ± 1.23	96.13 ± 1.89	94.27 ± 2.87	92.78 ± 1.98	88.58 ± 1.12	94.44 ± 0.89	93.34 ± 2.95	**96.14 ± 1.51**
5	99.32 ± 0.38	99.13 ± 0.81	99.03 ± 0.63	99.51 ± 0.52	99.18 ± 0.67	99.36 ± 0.28	99.02 ± 0.55	**99.48 ± 0.21**	99.46 ± 0.61
6	45.06 ± 5.21	72.82 ± 3.62	75.97 ± 2.38	76.30 ± 1.89	75.03 ± 2.76	78.47 ± 1.54	59.45 ± 3.87	85.21 ± 1.86	**98.71 ± 0.83**
7	77.42 ± 4.52	77.41 ± 3.68	78.64 ± 5.32	79.42 ± 3.21	74.58 ± 4.78	86.20 ± 3.12	85.88 ± 2.09	91.18 ± 1.87	**98.91 ± 1.22**
8	73.58 ± 1.12	69.63 ± 5.87	67.61 ± 3.51	59.94 ± 3.89	72.42 ± 3.21	82.26 ± 2.48	86.89 ± 2.17	86.43 ± 2.34	**90.75 ± 2.74**
9	95.44 ± 0.42	95.57 ± 2.31	96.81 ± 1.63	98.67 ± 1.38	94.67 ± 2.51	**99.92 ± 0.18**	99.36 ± 0.45	99.52 ± 0.21	99.61 ± 0.18
OA(%)	76.78 ± 2.01	77.23 ± 2.25	79.52 ± 1.74	81.52 ± 2.72	83.60 ± 2.29	85.12 ± 1.99	86.44 ± 1.97	89.75 ± 1.34	**91.08 ± 1.07**
AA(%)	80.08 ± 2.54	80.89 ± 2.74	82.29 ± 1.49	81.97 ± 1.23	83.21 ± 1.86	85.50 ± 1.78	85.86 ± 2.19	91.32 ± 1.86	**94.66 ± 0.91**
KC(%)	69.32 ± 2.52	70.25 ± 2.62	73.64 ± 2.01	76.10 ± 2.71	78.67 ± 2.73	80.59 ± 2.41	82.05 ± 1.73	85.52 ± 1.96	**88.50 ± 1.37**

proposed method exhibits stronger robustness and stability across different datasets.

2) Qualitative Results: To provide a clearer visual comparison of the classification performance of different methods, we present the classification maps generated on the IP, SA, and PU datasets, along with their corresponding ground-truth labels. As shown in Figs. 14–16, the classification maps generated by the proposed MGPDO method are the closest to the ground-truth labels.

Specifically, the highlighted regions *A* in Figs. 14–16 contain numerous structurally complex samples due to the diversity of land-cover types, and the heterogeneous pixels in

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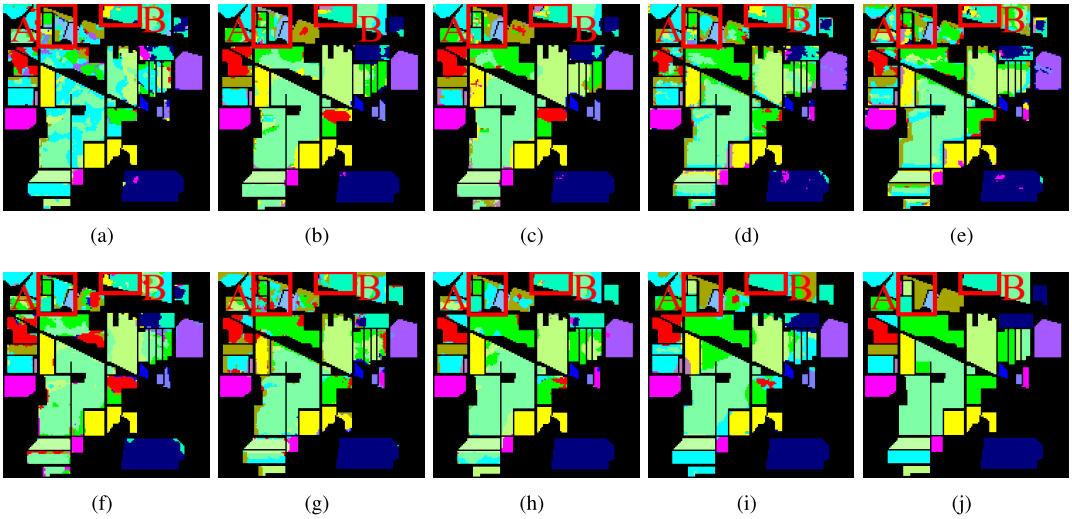


Fig. 14. Classification maps of the IP dataset. Some regions where the MGPDO yields better classification maps have been highlighted within the red box. (a) SSRN. (b) MCNN. (c) SPRN. (d) DCFSL. (e) Gia-CFSL. (f) ADAFSL. (g) RPCL-FSL. (h) CASCL. (i) MGPDO. (j) Ground truth.

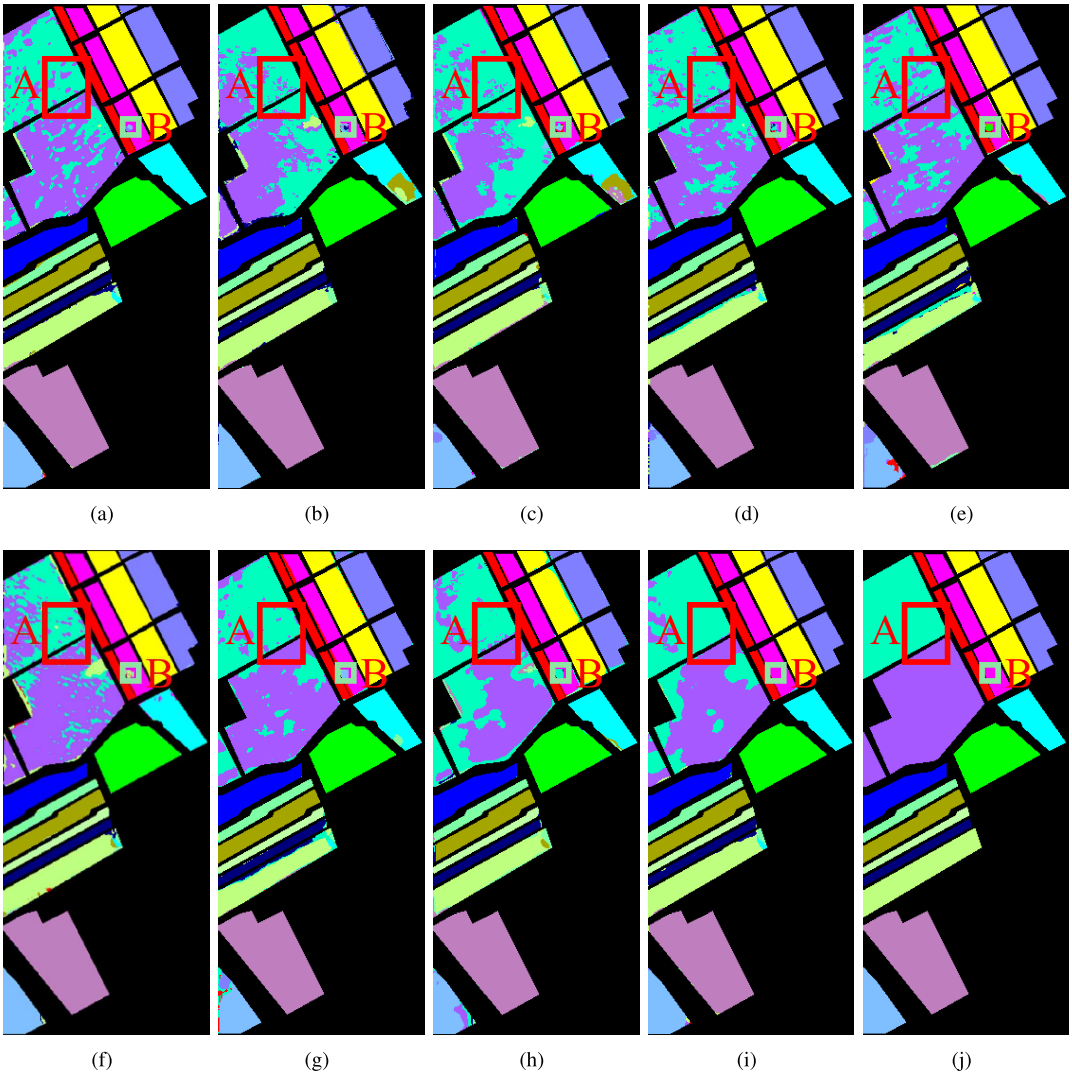


Fig. 15. Classification maps of the SA dataset. Some regions where the MGPDO yields better classification maps have been highlighted within the red box. (a) SSRN. (b) MCNN. (c) SPRN. (d) DCFSL. (e) Gia-CFSL. (f) ADAFSL. (g) RPCL-FSL. (h) CASCL. (i) MGPDO. (j) Ground truth.

these samples significantly interfere with feature extraction. In these figures, methods such as ADAFSL, RPCL-FSL, and CASCL exhibit substantial misclassification in these regions, particularly for the tenth class “soybean notill” and the 12th

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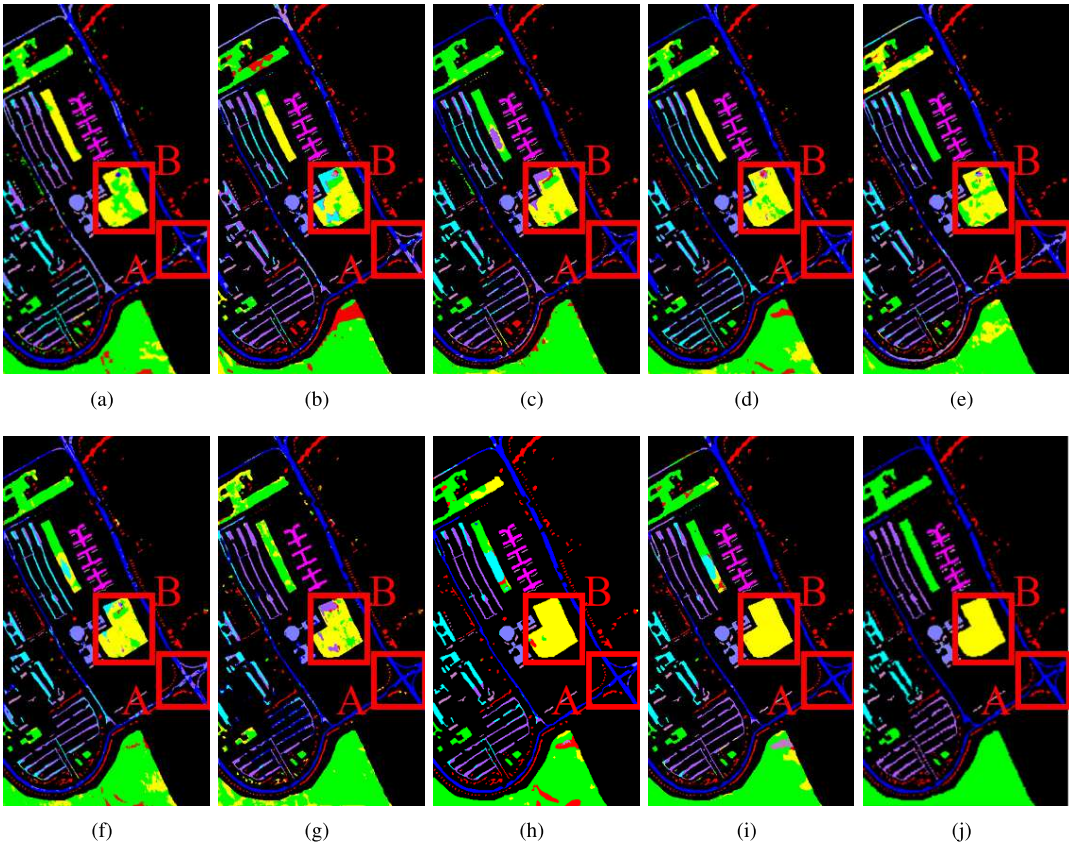


Fig. 16. Classification maps of the PU dataset. Some regions where the MGPDO yields better classification maps have been highlighted within the red box. (a) SSRN. (b) MCNN. (c) SPRN. (d) DCFSL. (e) Gia-CFSL. (f) ADAFSL. (g) RPCL-FSL. (h) CASCL. (i) MGPDO. (j) Ground truth.

class "soybean clean" in IP, the eighth class "grapes untrained" and the 15th class "Vinyard untrained" in SA, and the ninth class "shadows" in PU. In contrast, MGPDO achieves accurate classification in these challenging regions *A*, primarily due to its MSFE module, which effectively integrates fine-grained and global contextual information to learn consistent and discriminative features. Moreover, the highlighted regions *B* in Figs. 14–16 involve categories with the high intraclass diversity, such as the 15th class "buildings grass trees drives" in IP, the 16th class "Vinyard untrained trellis" in SA, and the fourth class "trees" in PU. As shown, the diverse feature distributions within these classes are difficult to capture with limited labeled samples, leading prototype-based methods such as DCFSL, RPCL-FSL, and CASCL to frequent misclassifications within these classes. In comparison, MGPDO constructs multiple Gaussian prototypes for each class, enabling accurate modeling of complex and diverse intraclass feature distributions. As a result, the classification map of the proposed MGPDO method is the closest to the ground truth.

In addition, in order to visually demonstrate the feature extraction performance of different methods, t-distributed stochastic neighbor embedding (t-SNE) is applied to project the features extracted by three cross-domain few-shot learning methods (DCFSL, CASCL, and MGPDO) into a 2-D space for the IP, SA, and PU datasets, as shown in Fig. 17. Compared with DCFSL and CASCL, the proposed method MGPDO achieves larger interclass feature distances and smaller intraclass distances on these datasets, demonstrating clearer classification boundaries and indicating that the learned features are more separable. To highlight the improvements more clearly, significantly enhanced regions are marked with red boxes in Fig. 17. Specifically, on the IP dataset [see Fig. 17(a)–(c)], the tenth class "soybean notill", the 11th class "soybean mintill", and the 12th class "soybean clean" exhibit similar land-cover types, which results in noticeable feature confusion and unclear class boundaries in the feature spaces of DCFSL and CASCL methods. In contrast, MGPDO effectively enlarges the interclass distances while reducing intraclass variability, thereby minimizing class confusion and improving overall separability. On the SA dataset [see Fig. 17(d)–(f)], the eighth class "grapes untrained" and the 15th class "vineyard untrained" show significant overlap in the feature spaces of DCFSL and CASCL. However, MGPDO enhances feature discrimination by increasing the interclass distance. In addition, the seventh class "celery" and the tenth class "corn senesced green weeds" display tighter intraclass distances under MGPDO, reflecting better intraclass consistency. On the PU dataset [see Fig. 17(g)–(i)], the third class "gravel" and the eighth class "self-blocking" are misclassified as a single cluster by DCFSL and CASCL due to their similar spatial–spectral characteristics. However, MGPDO increases its feature distance, forming two distinct clusters in the feature space. Moreover, the second class "meadows" and the sixth class "bare soil" exhibit more compact intraclass distributions under MGPDO, further

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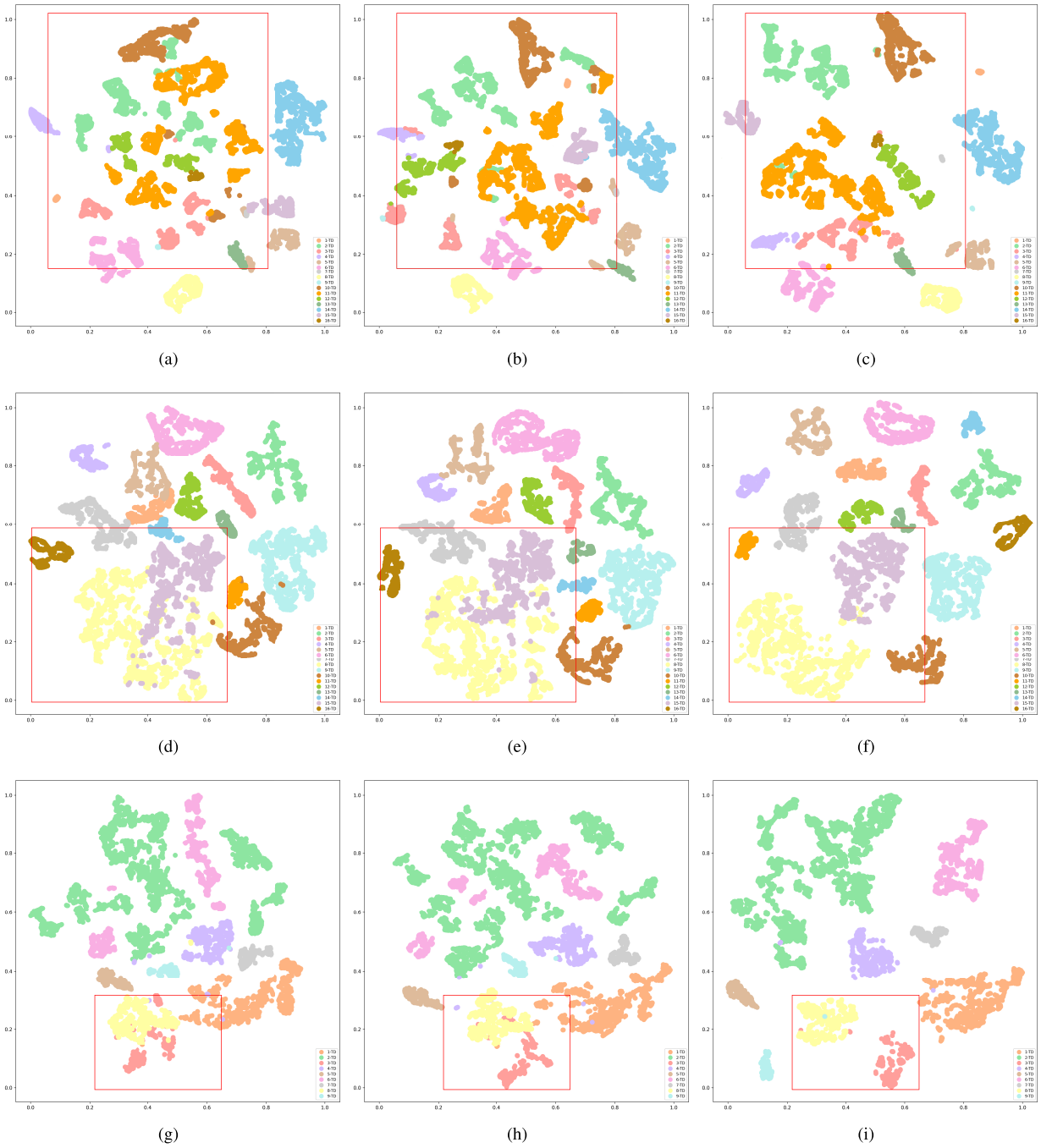


Fig. 17. 2-D feature visualization on three datasets. (a) DCFSL features on IP. (b) CASCL features on IP. (c) MGPDO features on IP. (d) DCFSL features on SA. (e) CASCL features on SA. (f) MGPDO features on SA. (g) DCFSL features on PU. (h) CASCL features on PU. (i) MGPDO features on PU.

enhancing overall separability. In summary, as shown in Fig. 17(c), (f), and (i), MGPDO achieves both compact intraclass distributions and clear interclass separations in the feature space. These results demonstrate the effectiveness of the CDTL module in reducing intraclass variation and enlarging interclass distances, which significantly enhances the separability of the learned features.

E. Computational Complexity

To assess the computational efficiency of different FSL methods, we compare training time, testing time, floating point operations (FLOPs), and the number of parameters under a five-shot setting on the target-domain datasets, as summarized in Table X. All experiments are conducted under identical hardware configurations to ensure fair comparison. For cross-domain methods (DCFSL, Gia-CFSL, ADAFSL, RPCL-FSL, CASCL, and our MGPDO), training time primarily reflects the transfer learning, while testing time corresponds to the inference time on unlabeled target samples. FLOPs and parameter counts are further reported to characterize model complexity. As shown in Table X, MGPDO provides competitive training and inference efficiency while maintaining reasonable model complexity.

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TABLE X
COMPUTATION TIME (S), FLOPS (M), AND PARAMETERS (M) ON EACH DATASET WITH DIFFERENT METHODS

	Methods	DCFSL	Gia-CFSL	ADAFSL	RPCL-FSL	CASCL	Ours
IP	Training time	2018.61	3192.76	2763.12	1429.74	1857.34	2065.82
	Testing time	1.58	1.21	2.63	1.32	0.98	1.68
	FLOPs	45.13	125.21	39.23	46.31	32.14	48.32
	#Params	3.63	2.23	2.02	3.82	0.61	3.87
SA	Training time	2077.15	4362.88	2348.53	1446.75	1775.36	2165.10
	Testing time	6.34	6.13	12.31	5.69	5.67	6.61
	FLOPs	45.13	125.20	39.24	46.31	32.14	48.33
	#Params	3.62	2.21	2.03	3.83	0.61	3.86
PU	Training time	1342.58	1967.32	1424.21	966.48	1330.75	1395.35
	Testing time	6.12	4.13	8.34	3.15	2.62	6.31
	FLOPs	45.14	125.21	39.23	46.32	32.13	48.33
	#Params	3.63	2.21	2.03	3.82	0.60	3.87

TABLE XI
ABLATION STUDY OF MGPDO: CONTRIBUTION OF FEATURE, ALIGNMENT, AND METRIC MODULES (OA %)

Feature		Alignment		Metric		Dataset		
DCFSL	MSFE	CDTL(SD)	CDTL(TD)	Gaussian	Multi-Proto	IP	SA	PU
						58.52	76.13	68.56
✓						66.81	89.34	81.32
✓	✓					75.04	91.95	86.22
✓	✓	✓				77.15	92.34	86.85
✓	✓	✓	✓			79.24	93.13	87.48
✓	✓			✓		76.83	92.07	88.31
✓	✓			✓	✓	77.25	92.52	90.17
✓	✓	✓	✓	✓		80.51	93.82	90.55
✓	✓	✓	✓	✓	✓	81.32	94.19	91.08

F. Classification Accuracies With Different Numbers of Labeled Samples

To analyze the influence of different numbers of labeled training samples on the classification performance of CNN-based methods and several FSL-based methods, we randomly selected the number of labeled samples per class from $\{1, 2, 3, 4, 5\}$, and the final classification results on three datasets are reported in Fig. 18. The classification performance of the various methods declines as the number of training samples decreases, which is consistent with expectations. However, the proposed MGPDO consistently achieves higher classification accuracy compared with other methods, even when the number of training samples is extremely limited (e.g., one sample per class).

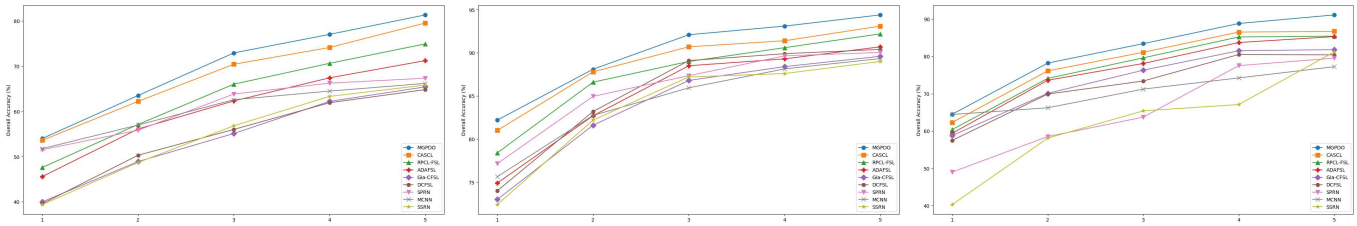
G. Ablation Study

The proposed MGPDO framework consists of six key modules, including MSFE, dual-level alignment (DCFSL and CDTL with source/target-domain anchors), and metric learning components (Gaussian prototype and multiprototype modeling). To better analyze the roles of these components, we conducted ablation experiments to evaluate their contributions. As shown in Table XI, we focused on nine scenarios: 1) only limited target-domain samples without pretraining, adaptation or prototype-based classification; 2) adds the DCFSL module for domain-level alignment via conditional adversarial loss with a prototypical metric; 3) adds the MSFE module to enhance MSFE based on 2); 4) adds the CDTL(SD) module using source-domain prototypes as anchors based on 3); 5) adds the CDTL(TD) module with target-domain anchors based on 4); 6) replaces the prototypical metric with a Gaussian prototype metric (GPM) based on 3); 7) introduces multiprototype modeling to capture intraclass diversity on 6); 8) adds the full feature and alignment modules (MSFE, CDTL) based on 6); and 9) combines all modules-MSFE, CDTL (SD/TD), Gaussian metric, and multiprototype modeling-based on 8), forming the complete MGPDO framework.

As shown in Table XI, we conduct a stepwise ablation study to assess the individual contributions of each component in the MGPDO framework: In scenario 1), the model is trained solely on limited target-domain samples without adaptation or prototype-based supervision. In scenario 2), introducing the DCFSL module with conditional adversarial alignment and prototypical metric improves OA by 8.29%, 13.21%, and 12.76% on IP, SA, and PU, respectively. In scenario 3), incorporating MSFE enhances multiscale spatial–spectral representation, yielding further gains of 8.23%, 2.61%, and 4.9%. In scenario 4), CDTL(SD) is added to perform class-level alignment using source-domain anchors, resulting in additional improvements of 2.11%, 0.39%, and 0.63%. In scenario 5), target-domain anchors are introduced in CDTL(TD) for cyclic alignment, providing further OA increases of 2.09%, 0.79%, and 0.63%. In scenario 6), the prototypical metric is replaced with a GPM, improving boundary modeling with gains of 1.79%, 0.12%, and 2.09%. In scenario 7), a multiprototype strategy is employed to capture intraclass variability, leading to additional gains of 0.42%, 0.45%, and 1.86%. In scenario 8), all feature and alignment modules (MSFE + CDTL) are jointly integrated, further enhancing performance by 3.68%, 1.75%, and 2.24%. In scenario 9), the complete MGPDO framework, comprising MSFE, CDTL (SD/TD), Gaussian metric, and multiprototype modeling, achieves the highest OA across all datasets (IP: 81.32%, SA: 94.19%, and PU: 91.08%), validating its effectiveness in addressing both global and fine-grained domain shifts.

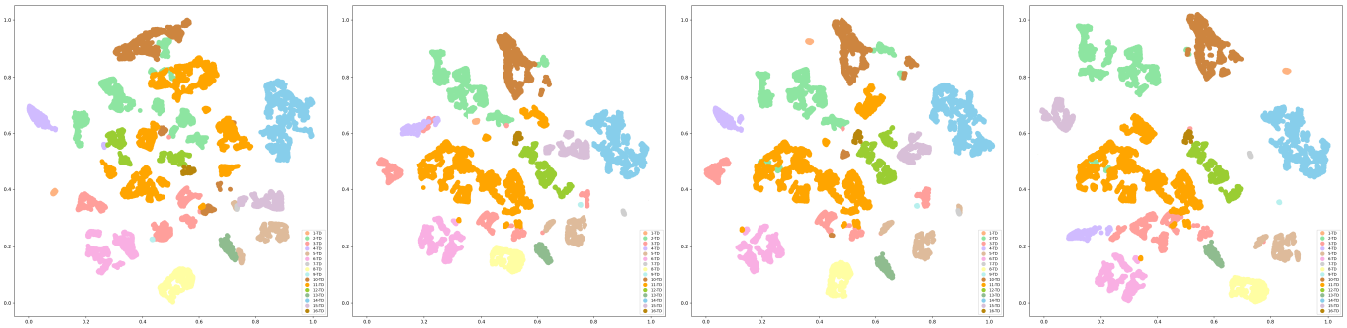
To further illustrate the progressive effect of each module in MGPDO, we provide t-SNE visualizations under four representative scenarios on the IP dataset, as shown in Fig. 19. Each subfigure corresponds to a different stage of module integration.

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(a) (b) (c)

Fig. 18. OA (%) of all methods under different numbers of labeled samples per class in the target-domain dataset: (a) IP, (b) SA, and (c) PU.



(a) (b) (c) (d)

Fig. 19. t-SNE visualizations of feature distributions under different module configurations on the IP dataset. (a) DCFSL, (b) DCFSL + MSFE, (c) DCFSL + MSFE + CDTL, and (d) DCFSL + MSFE + CDTL + MGPM.

1) Represents the baseline model with only the DCFSL for domain-level alignment via conditional adversarial loss and a prototypical metric. The feature space remains dispersed with considerable class overlap.
2) Adds the MSFE module to enhance multiscale spatial–spectral representation, yielding more compact and coherent class distributions.
3) Further incorporates the CDTL module for class-level alignment through dual-anchor triplet losses, leading to clearer cluster separation across domains.
4) Completes the framework by introducing the MGPM module for multiprototype modeling, which captures intraclass variation and sharpens class boundaries. These results visually confirm the effectiveness of each proposed module in progressively enhancing the quality of feature representations on the IP dataset.

IV. Conclusion

In this article, we propose the MGPDO method for cross-domain few-shot HSI classification. Compared with existing methods that mainly focus on global domain alignment and prototype-based metric learning, MGPDO simultaneously optimizes both interclass alignment and intraclass feature modeling, thereby achieving more accurate cross-domain classification. Specifically, we introduce a CDTL, which enhances the similarity between positive prototype pairs while suppressing the correlation of negative pairs to achieve interclass alignment across domains. Moreover, to more accurately model the complex intraclass feature distributions, we propose an MGPM module, which leverages multiple Gaussian prototypes per class to capture intraclass diversity. In recent years, multimodal learning has shown significant advantages in enhancing model generalization by leveraging complementary information from multiple modalities, thereby offering a promising direction for cross-domain few-shot classification. However, in multimodal scenarios, it is not only necessary to align the source and target domains, but also essential to address the discrepancies across different modalities. To address these challenges, we intend to extend the proposed method to multimodal scenarios in future work, aiming to design a unified optimization strategy that jointly addresses both domain alignment and modality alignment to further improve the model's accuracy in multimodal cross-domain few-shot tasks.

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